

Recommendations Report

Executive Summary

This report presents recommendations based on the results obtained from the Logistic Regression and Random Forest models developed for predicting subscriber churn. The objective of the project was to identify customers who are likely to cancel their subscriptions and provide insights that can support customer retention initiatives. The results show that the Random Forest model significantly outperformed the Logistic Regression model and is therefore recommended as the preferred solution. The findings also confirm that customer engagement and platform activity are the strongest drivers of churn, which aligns with the challenges identified in the STADIOchoice business case and the literature reviewed in Part A.

Which Model Is Best Between Model 1 and Model 2?

Based on the performance results, the Random Forest model is the preferred model for predicting subscriber churn. The model achieved an accuracy of 97.6%, precision of 97.4%, recall of 97.8%, F1-score of 97.6%, and a ROC-AUC score of 99.7%. These results demonstrate an excellent ability to correctly identify both churned and retained subscribers. Although the Logistic Regression model achieved a very high recall score of 99.4%, it produced an accuracy of only 64.7% and a precision score of 58.8%. The confusion matrix showed that the model incorrectly classified 350 retained subscribers as churned. This indicates that the model was overly sensitive and tended to overpredict churn. Such a model would likely result in unnecessary retention campaigns and increased costs because many loyal customers would be incorrectly targeted as churn risks. The Random Forest model produced only 24 incorrect predictions out of 1,000 observations and achieved substantially better results across every evaluation metric. These findings suggest that customer churn is influenced by complex and non-linear relationships that are better captured by ensemble learning techniques. For this reason, the Random Forest model is recommended as the final model for deployment.

How Can the Selected Model Be Improved?

Although the Random Forest model performed very well, there are several ways it can be improved. First, the model settings can be fine-tuned by testing different values for parameters such as the number of trees and tree depth. This may help improve accuracy and reduce the remaining prediction errors. Second, more customer behaviour data can be added to the model. Information such as payment failures, customer support interactions, search activity, retention offers, content completion rates, and viewing frequency may help the model identify churn patterns more accurately. Explainable AI techniques such as SHAP can also be used to show why customers are predicted to churn, making the model easier for management to understand and trust. Finally, the model should be retrained regularly to ensure it stays accurate as customer behaviour changes over time.

How Will the Selected Model Need to Be Adapted to Work with the Requested Data from SS1?

The Netflix dataset provided a useful environment for building and testing the model. However, the data requested from STADIOchoice in SS1 is considerably richer and would require the model to be adapted. The requested datasets include subscriber information, billing history, streaming activity, retention records, and customer support interactions. These datasets would first need to be integrated using a common Household ID or Subscriber ID before preprocessing and feature engineering can begin. Additional features could then be engineered from the available data. Examples include the number of failed payments, support contact frequency, search activity, content diversity scores, accepted retention offers, viewing completion rates, and subscription tenure. Since the current study found that engagement variables such as Watch Hours, Last Login Days, Recency Score, and Low Engagement were the most important predictors of churn, the additional behavioural data available within STADIOchoice is expected to further improve prediction performance. The modelling process itself would remain broadly similar. Preprocessing, feature engineering, and classification modelling would still be required. However, the larger and more detailed STADIOchoice datasets would allow the Random Forest model to generate more accurate and business-specific churn predictions. This approach ensures that the final implementation remains consistent with the methods developed in this study while taking full advantage of STADIOchoice's operational data.

Do the Results Align With the Related Literature From Part A?

The findings of this study align strongly with the literature reviewed in Part A. The study *Churn Buster: Uncovering Patterns and Predicting Churn in OTT Platforms* found that customer engagement and behavioural activity are important predictors of churn. The results of the current study support this conclusion because variables such as Watch Hours, Last Login Days, Recency Score, and Low Engagement were identified as the strongest predictors of subscriber churn.

The study *A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI* highlighted the value of advanced machine learning techniques for customer retention. The strong performance of the Random Forest model supports this finding and demonstrates that machine learning models are effective tools for identifying subscribers at risk of cancelling their subscriptions.

The study *Customer Churn and Content Strategy Analysis for Netflix* reported that viewing behaviour and engagement metrics were among the most important factors influencing churn. Similar results were obtained in this project, where Watch Hours emerged as the most important feature and several other engagement-related variables ranked highly in the Random Forest feature importance analysis.

Some differences between the current study and the literature should also be acknowledged. The studies reviewed in Part A used larger datasets, more advanced algorithms such as XGBoost, CatBoost and AdaBoost, and additional techniques such as SHAP analysis. The current study was limited to the features available within the Netflix Customer Churn Dataset and focused on Logistic Regression and Random Forest models. Despite these differences, both the literature and the current study consistently identified customer engagement and viewing behaviour as the strongest drivers of churn. This consistency increases confidence in the validity and reliability of the findings.

Conclusion

The Random Forest model is recommended for STADIOchoice because it significantly outperformed the Logistic Regression model across all evaluation metrics and demonstrated a strong ability to identify both churned and retained subscribers. The analysis showed that customer engagement and platform activity are the strongest drivers of churn, with Watch Hours, Last Login Days, Recency Score, Low Engagement, and High Engagement emerging as the most important predictors. The results are consistent with the findings from the literature review, suggesting that implementing the Random Forest model with STADIOchoice's richer customer data could improve churn prediction, strengthen retention strategies, and support long-term streaming profitability.